

INFORMATION TECHNOLOGY: PRACTICAL EXAMINATION

INSTRUCTIONS TO TEACHERS

1. For the examination requirements for Information Technology please read pages 3 – 5 of the *Subject Assessment Guidelines*.
2. The school will receive a data disk containing files for the examination. These should be checked by the Information Technology teacher two days before the examination. Please refer to the instructions concerning data files supplied on page 3.
3. Do not open the sealed envelopes containing the examination papers and marking guidelines until the day of the examination.
4. Each candidate must be supplied with a disk (or space on a network drive) for his/her exclusive use throughout the examination. No candidate may have access to another candidate's work during the examination. Disable any shared folders on the network and ensure that the candidates cannot use their own flash drives or memory sticks. If your candidates normally use usernames and passwords then create a new username for each candidate.
5. Please ensure that the machines have been checked and all extraneous data/programs have been removed. Disable communications programs that will allow chatting or messaging. Ensure that no communication can take place between machines. No candidate may access the Internet during the examination.
6. Information Technology teachers are allowed to invigilate the Information Technology examinations, in case there are problems with the hardware and system software. Obviously, they may not assist the candidates in any way with their particular programme(s).
7. The candidates may use the help facility that has been supplied with the programming language's software. They may not use any other manuals or documentation. In the case of Java, they may use the standard APIs available on the java.sun.com site. These should be downloaded **before** the exam and copied onto each candidate's disk space provided for the examination.
8. The three hours of the examination does not include printing time. It is exclusively for coding. At the end of the three hours, printing must be done either by the candidates, or if conditions are not suitable for this, by the Information Technology teacher. No code or data may be altered after the three-hour time period has expired.

9. Teachers must inform their candidates what printing arrangements have been made at their school.
10. Information Technology teachers must mark a sample of five papers of their candidates' work, using the supplied interim memorandum and marksheet as a guideline. They must then meet with their cluster group, as outlined in the *Subject Assessment Guidelines* document.
11. The mark scheme and solution given is just one way of solving the problem. If your candidate has solved the problem in a different manner that still uses good programming techniques and incorporates all aspects of the question, then full marks may be awarded.
12. Place **all** your candidate's printed work and updated marksheets in a **single** envelope with the school name and examination number clearly marked on the outside of the envelope. **DO NOT PLACE CANDIDATES' WORK IN INDIVIDUAL ENVELOPES.** Unlike cluster moderation, you must submit every candidate's practical examination.

An electronic copy of all the candidates' examinations and their spreadsheet mark schedule should be provided on a single CD. In the case of their examination programs, a copy of the entire folder with all files used should be put onto the CD. The teacher may not delete or make any changes to these files under any circumstances, even if the files do not affect the outcome of the programme/solution.

13. The interim mark scheme may not be used to mark the candidates' exams. The updated mark scheme must be obtained from <http://groups.yahoo.com/group/iebcompza> or www.ieb.co.za one week after the examination. A candidate's work must be marked using the mark scheme spreadsheet so that the spreadsheet will calculate the candidate's total mark. Place a printed copy of the spreadsheet in front of each candidate's solution. Make sure the examination number is clearly visible on each marked script. Teachers must show evidence that the programme print-outs have been marked by indicating ticks and so on. Marking should be done in red pen. Any comments and justifications are helpful for the moderators.
14. You will receive a summary mark schedule for the centre which must be returned with the candidates' marksheets and printouts. These marks should also be entered via the IEB online facility.
15. Check that the code given will successfully connect to the database that you are using (MS Access or MySQL). Candidates who are using other databases will need to have the database converted. You may need to alter the code to perform the connection, however no other code may be changed and any changes need to be submitted to the IEB together with each candidate's marked script by 15 November 2009.

TO THE TEACHER RESPONSIBLE FOR CANDIDATES IN THE INFORMATION TECHNOLOGY PRACTICAL EXAMINATION**INSTRUCTIONS CONCERNING DATA FILES SUPPLIED ON DISK**

You have been supplied with files on a CD. It is the responsibility of the Information Technology teacher to check **two days before the examination** that they can open the files and that the files can be read by the relevant applications. Printouts of the files are given, starting on page 5 in this document. The details of the files are as follows:

Files for Learners: Please consult the folders on the CD. These are the important files only.

Question 1:**Java (with Access, Text-based/Netbeans):**

FindAPet.java
pets.mdb

Java (with MySQL, Netbeans):

FindAPet.java
pets.sql

Delphi (with Access)

Question1.pas
Pets.mdb
Question1.dpr

Question 2:**Java (Text-based)**

Conference.txt
Delegate.java
ConventionApp.java

Java (A Netbeans project is provided, ensure it works!)

Conference.txt
Src/convention/Delegate.java
Src/convention/ConventionApp.java

Delphi

Conference.txt
uDelegate.pas
ufrmMain.pas
ConventionApp.dpr

If you have any problems with accessing the files on disk, please notify the IEB immediately.

Those using the Netbeans project for question 2 will require Netbeans 6.5.1 in order to open this. **Should the centre not have this, the teacher may create his/her own project by copying the files in the 'SRC' folder within the project folder and the *conference.txt* file into a new project. As a last resort a teacher may create a new form provided it does nothing more than the text-based menu system – that is provide the menu options with no commands in it.**

If a teacher changes the project for any reason a full print-out and electronic copy must be provided with a letter explaining the need for the change and which is countersigned by the School Head. Teachers are strongly advised against doing this unless viewed as entirely necessary. Doing this does put all the candidates' results under careful scrutiny.

Individual copies of these files will be required by each candidate writing the Information Technology Practical Examination. Please copy the relevant files onto a disk for each candidate's exclusive use, or onto the disk space that will be used exclusively by each candidate.

Teachers are advised to supply only those files that are needed for the pupils in their centre (don't supply the MySQL files if your candidates are writing with Access, or the Java files if they are writing Delphi).

Confidentiality is of the utmost importance. Under no circumstances should candidates have access to these files before the examination.

After the examination, create a folder for each candidate using their name and examination number. Copy ALL the candidate's files into their particular folder. Be sure to include the data files that are accessed by the programme. Ensure that the candidates have not used path names for the data files in their programs. Send the CD with the marked practical exams to the IEB. Ensure that you have attended the cluster group meetings and have used the updated mark schemes to mark all the practical exams.

Files for Learners

The following pages contain the printouts of the code supplied to each of the candidates. Copies of these files are on the CD that is provided to schools. They are provided here in case of emergency and for your reference.

Student Files: Question 1

File: FindAPet.java (For Access)

```
import java.sql.*;
import javax.swing.*;

public class FindAPet
{
    private Connection conn;

    public FindAPet ()
    {
        try
        {
            Class.forName ("sun.jdbc.odbc.JdbcOdbcDriver");
            conn = DriverManager.getConnection ("jdbc:odbc:DRIVER={Microsoft
                Access Driver (*.mdb)};DBQ=pets.mdb");
            System.out.println ("Connection successful");
        }
        catch (Exception e)
        {
            System.out.println ("ERROR: " + e);
            e.printStackTrace ();
        }

        char choice;
        do
        {
            choice = getMenuChoice ();

            if (choice == 'A')
            {

            }

            else if (choice == 'B')
            {

            }

            else if (choice == 'C')
            {

            }

            else if (choice == 'D')
            {

            }

            else if (choice == 'E')
            {

            }

        }
        while (choice != 'X');
    }
}
```

```

    public static void main (String[] args)
    {
        new FindAPet ();
    }

    public char getMenuChoice ()
    {
        String menu = "A) Display all owners, sorted by name\nB) Display
                      residents of a particular city\nC) Display
                      pet type statistics\nD) Show pet type
                      owners\nE) Add a new pet\nX) Exit";

        char choice = JOptionPane.showInputDialog (menu).toUpperCase ().charAt
                                (0);

        return choice;
    }
}

```

File: FindAPet.java (For MySQL)

```

import java.sql.*;
import javax.swing.*;

public class FindAPet
{
    private static final String driver = "com.mysql.jdbc.Driver";
    private static final String url = "jdbc:mysql://localhost:3306/Pets";
    private static final String user = "root";
    private static final String pass = "nbuser";
    private Connection conn;

    public FindAPet ()
    {
        try {
            Class.forName(driver);
            conn = DriverManager.getConnection(url, user, pass);
        }
        catch (ClassNotFoundException ex) {
            System.out.println("Class not found");
        }
        catch (SQLException ex) {
            System.out.println("SQL error");
        }
    }

    char choice;
    do
    {
        choice = getMenuChoice ();

        if (choice == 'A')
        {

        }

        else if (choice == 'B')
        {

        }

        else if (choice == 'C')
        {

```

```

    }

    else if (choice == 'D')
    {

    }

    else if (choice == 'E')
    {

    }
}
while (choice != 'X');
}

public static void main (String[] args)
{
    new FindAPet ();
}

public char getMenuChoice ()
{
    String menu = "A) Display all owners, sorted by name\nB) Display
                  residents of a particular city\nC) Display
                  pet type statistics\nD) Show pet type
                  owners\nE) Add a new pet\nX) Exit";

    char choice = JOptionPane.showInputDialog (menu).toUpperCase ().charAt
                                (0);

    return choice;
}
}

```

File: Question1.pas

```

unit Question1;

interface

uses
    Windows, Messages, SysUtils, Variants, Classes, Graphics, Controls, Forms,
    Dialogs, DB, ADODB, Menus;

type
    TForm1 = class(TForm)
        qry1: TADOQuery;
        mm1: TMainMenu;
        Queries1: TMenuItem;
        OwnersSortedbyLastnameFirstname1: TMenuItem;
        Allownersforacity1: TMenuItem;
        Countpettypes1: TMenuItem;
        Ownersforaspecificpettype1: TMenuItem;
        Addapet1: TMenuItem;
    private
        { Private declarations }
    public
        { Public declarations }
    end;

var
    Form1: TForm1;

```

implementation

```
{ $R *.dfm }
```

end.

File: Pets.sql (for MySQL)

```
-- MySQL Administrator dump 1.4
```

```
--
```

```
-- Server version 5.1.33-community
```

```
/*!40101 SET @OLD_CHARACTER_SET_CLIENT=@@CHARACTER_SET_CLIENT */;
```

```
/*!40101 SET @OLD_CHARACTER_SET_RESULTS=@@CHARACTER_SET_RESULTS */;
```

```
/*!40101 SET @OLD_COLLATION_CONNECTION=@@COLLATION_CONNECTION */;
```

```
/*!40101 SET NAMES utf8 */;
```

```
/*!40014 SET @OLD_UNIQUE_CHECKS=@@UNIQUE_CHECKS, UNIQUE_CHECKS=0 */;
```

```
/*!40014 SET @OLD_FOREIGN_KEY_CHECKS=@@FOREIGN_KEY_CHECKS, FOREIGN_KEY_CHECKS=0 */;
```

```
/*!40101 SET @OLD_SQL_MODE=@@SQL_MODE, SQL_MODE='NO_AUTO_VALUE_ON_ZERO' */;
```

```
--
```

```
-- Create schema pets
```

```
--
```

```
CREATE DATABASE IF NOT EXISTS pets;
```

```
USE pets;
```

```
--
```

```
-- Definition of table `tblowners`
```

```
--
```

```
DROP TABLE IF EXISTS `tblowners`;
```

```
CREATE TABLE `tblowners` (
  `OwnerID` int(10) NOT NULL AUTO_INCREMENT,
  `LastName` varchar(255) DEFAULT NULL,
  `FirstName` varchar(255) DEFAULT NULL,
  `Telephone` varchar(255) DEFAULT NULL,
  `City` varchar(255) DEFAULT NULL,
  PRIMARY KEY (`OwnerID`)
) ENGINE=InnoDB AUTO_INCREMENT=53 DEFAULT CHARSET=latin1;
```

```
--
```

```
-- Dumping data for table `tblowners`
```

```
--
```

```
/*!40000 ALTER TABLE `tblowners` DISABLE KEYS */;
```

```
INSERT INTO `tblowners` (`OwnerID`, `LastName`, `FirstName`, `Telephone`, `City`)
VALUES
```

```
(5, 'Schutte', 'Monique', '074 521 8820', 'Cape Town'),
(6, 'Fouche', 'Emma', '082 221 2659', 'Bloemfontein'),
(7, 'Schutte', 'Nicole', '072 393 5538', 'Centurion'),
(8, 'Ngubane', 'Tanya', '082 658 3903', 'Polokwane'),
(9, 'Bezuidenhout', 'Jana', '082 770 2752', 'Polokwane'),
(10, 'Davey', 'Kelly', '084 490 3274', 'Ulundi'),
(11, 'Theunis', 'Melissa', '082 497 7912', 'Cape Town'),
(12, 'Raphael', 'Courtney', '084 816 1243', 'Polokwane'),
(13, 'Watts', 'Michelle', '082 617 9619', 'Richards Bay'),
(14, 'Adriaans', 'Simone', '084 512 8971', 'Midrand'),
(15, 'Ngubane', 'Amy', '084 598 3540', 'Midrand'),
```

```
(16, 'Way', 'Nadia', '082 377 3329', 'Pretoria'),
(17, 'Swanepoel', 'Nicky', '084 584 0536', 'Johannesburg'),
(18, 'Swanepoel', 'Chantel', '082 714 1751', 'Cape Town'),
(19, 'Grobler', 'Zinhle', '074 299 3948', 'Kempton Park'),
(20, 'Du Preez', 'Caitlin', '074 215 7964', 'Knysna'),
(21, 'Petzer', 'Charlene', '082 517 9383', 'Ulundi'),
(22, 'Duma', 'Jodi', '082 827 3009', 'Cape Town'),
(23, 'Brophy', 'Palesa', '074 631 7922', 'Port Elizabeth'),
(24, 'Wagener', 'Lerato', '074 897 1259', 'Ulundi'),
(25, 'Watts', 'Jessica', '072 691 5542', 'Centurion'),
(26, 'Adriaans', 'Mikaela', '083 550 2341', 'Port Elizabeth'),
(27, 'West', 'Christine', '083 434 3770', 'Centurion'),
(28, 'Williams', 'Toni', '084 689 5760', 'Durban'),
(29, 'Bezuidenhout', 'Daniel', '084 456 9284', 'Midrand'),
(30, 'Mbeki', 'Nickens', '084 249 0903', 'Pretoria'),
(31, 'Theunis', 'Andrew', '074 355 8272', 'Cape Town'),
(32, 'Fouche', 'Kevin', '074 491 4832', 'East London'),
(33, 'Watts', 'David', '072 391 0409', 'Durban'),
(34, 'Kent', 'Junior', '084 391 1222', 'Cape Town'),
(35, 'Smit', 'Patrick', '084 644 7473', 'Port Elizabeth'),
(36, 'Theunis', 'Daouda', '073 243 1648', 'Midrand'),
(37, 'Fouche', 'Jacques', '073 331 0430', 'Johannesburg'),
(38, 'Moloto', 'Jarryd', '074 214 5504', 'Richards Bay'),
(39, 'Watts', 'Clinton', '072 360 8071', 'Pietermaritzburg'),
(40, 'Anfield', 'Thato', '073 788 0908', 'Centurion'),
(41, 'West', 'Peter', '072 581 0600', 'Richards Bay'),
(42, 'Van Wyk', 'Jacob', '083 473 2686', 'Cape Town'),
(43, 'Bezuidenhout', 'Lebogang', '083 874 0060', 'East London'),
(44, 'Petzer', 'Luigi', '074 575 5078', 'Port Elizabeth'),
(45, 'Ngubane', 'Chris', '083 470 8076', 'Knysna'),
(46, 'Theunis', 'Katleho', '083 798 4995', 'Ulundi'),
(47, 'Culling', 'Thabo', '072 564 6997', 'Richards Bay'),
(48, 'Louw', 'Mervin', '082 634 7456', 'Durban'),
(49, 'Way', 'Jonathan', '073 382 7666', 'Ulundi'),
(50, 'Moloto', 'Nhlanhla', '074 275 6930', 'Port Elizabeth'),
(51, 'Hlangani', 'Gift', '073 809 7287', 'Knysna'),
(52, 'Way', 'Johan', '083 561 8246', 'Richards Bay');
/*!40000 ALTER TABLE `tblowners` ENABLE KEYS */;
```

```
--
-- Definition of table `tblpets`
--
```

```
DROP TABLE IF EXISTS `tblpets`;
CREATE TABLE `tblpets` (
  `PetID` int(10) NOT NULL AUTO_INCREMENT,
  `PetName` varchar(255) DEFAULT NULL,
  `OwnerID` int(10) DEFAULT NULL,
  `PetType` varchar(255) DEFAULT NULL,
  `PetBreed` varchar(255) DEFAULT NULL,
  PRIMARY KEY (`PetID`),
  KEY `OwnerID` (`OwnerID`)
) ENGINE=InnoDB AUTO_INCREMENT=110 DEFAULT CHARSET=latin1;
```

```
--
-- Dumping data for table `tblpets`
--
```

```
/*!40000 ALTER TABLE `tblpets` DISABLE KEYS */;
INSERT INTO `tblpets` (`PetID`, `PetName`, `OwnerID`, `PetType`, `PetBreed`) VALUES
(11, 'Harley', 28, 'Cat', 'Abyssinian'),
(12, 'Cleo', 11, 'Bird', 'African Grey'),
(13, 'Buster', 40, 'Cat', 'Persian'),
```

(14, 'Donner', 11, 'Bird', 'Canary'),
 (15, 'Rusty', 14, 'Bird', 'Cockatoo'),
 (16, 'Maggie', 33, 'Cat', 'Siamese'),
 (17, 'Snoopy', 36, 'Bird', 'Parakeet'),
 (18, 'Paul', 51, 'Bird', 'Toucan'),
 (19, 'Tiger', 24, 'Dog', 'Doberman'),
 (20, 'Sparky', 12, 'Bird', 'Budgie'),
 (21, 'King', 43, 'Bird', 'Cockatoo'),
 (22, 'Casey', 34, 'Bird', 'Lovebird'),
 (23, 'Rocky', 37, 'Bird', 'Dove'),
 (24, 'Spot', 15, 'Dog', 'Boxer'),
 (25, 'Donner', 46, 'Cat', 'Burmese'),
 (26, 'Shelby', 32, 'Bird', 'Finch'),
 (27, 'Branford', 35, 'Cat', 'Burmese'),
 (28, 'King', 50, 'Dog', 'Beagle'),
 (29, 'Cody', 41, 'Cat', 'Abyssinian'),
 (30, 'Fluffy', 17, 'Bird', 'African Grey'),
 (31, 'Toby', 46, 'Bird', 'Toucan'),
 (32, 'Buster', 13, 'Cat', 'Burmese'),
 (33, 'Winston', 36, 'Dog', 'Yorkshire'),
 (34, 'Rex', 26, 'Bird', 'Cockatoo'),
 (35, 'Branford', 30, 'Dog', 'Rottweiler'),
 (36, 'Tiger', 28, 'Cat', 'Burmese'),
 (37, 'Prince', 39, 'Cat', 'Burmese'),
 (38, 'Molly', 40, 'Dog', 'Rottweiler'),
 (39, 'Murphy', 15, 'Cat', 'Burmese'),
 (40, 'Tiger', 42, 'Dog', 'Rottweiler'),
 (41, 'Charlie', 35, 'Dog', 'Pitbull'),
 (42, 'Winston', 52, 'Dog', 'Labrador'),
 (43, 'Sparky', 33, 'Cat', 'Abyssinian'),
 (44, 'Casey', 40, 'Bird', 'Canary'),
 (45, 'Cody', 23, 'Cat', 'Abyssinian'),
 (46, 'Sadie', 22, 'Dog', 'Chihuahua'),
 (47, 'Branford', 33, 'Dog', 'Golden Retriever'),
 (48, 'Charlie', 46, 'Bird', 'Dove'),
 (49, 'Shelby', 7, 'Bird', 'Lovebird'),
 (50, 'Charlie', 21, 'Cat', 'Persian'),
 (51, 'Bailey', 22, 'Bird', 'Parakeet'),
 (52, 'Rover', 44, 'Cat', 'Burmese'),
 (53, 'Dopey', 38, 'Cat', 'Burmese'),
 (54, 'Shadow', 51, 'Dog', 'Doberman'),
 (55, 'Rex', 26, 'Bird', 'Toucan'),
 (56, 'Sparky', 8, 'Cat', 'Persian'),
 (57, 'Toby', 19, 'Bird', 'Dove'),
 (58, 'Charlie', 47, 'Cat', 'Siamese'),
 (59, 'Jack', 44, 'Cat', 'Siamese'),
 (60, 'Rex', 47, 'Bird', 'African Grey'),
 (61, 'Fluffy', 16, 'Dog', 'Doberman'),
 (62, 'Lucy', 31, 'Dog', 'Rottweiler'),
 (63, 'Rusty', 17, 'Dog', 'Doberman'),
 (64, 'Daisy', 52, 'Cat', 'Siamese'),
 (65, 'Bailey', 24, 'Bird', 'Lovebird'),
 (66, 'Rocky', 25, 'Bird', 'Dove'),
 (67, 'Bailey', 41, 'Bird', 'Canary'),
 (68, 'Tiger', 30, 'Dog', 'Pitbull'),
 (69, 'Princess', 32, 'Dog', 'German Shepherd'),
 (70, 'Duke', 40, 'Cat', 'Persian'),
 (71, 'Princess', 15, 'Dog', 'Boxer'),
 (72, 'Spike', 22, 'Dog', 'German Shepherd'),
 (73, 'Harley', 37, 'Bird', 'Conure'),
 (74, 'Sadie', 49, 'Dog', 'Yorkshire'),
 (75, 'Paul', 6, 'Bird', 'Canary'),
 (76, 'Donner', 51, 'Bird', 'Finch'),
 (77, 'Barney', 29, 'Dog', 'Jack Russel'),

```

(78, 'Cody', 17, 'Bird', 'Toucan'),
(79, 'Dopey', 39, 'Dog', 'Doberman'),
(80, 'Molly', 38, 'Dog', 'German Shepherd'),
(81, 'Spot', 38, 'Cat', 'Persian'),
(82, 'Duke', 29, 'Dog', 'Beagle'),
(83, 'Rocky', 8, 'Cat', 'Siamese'),
(84, 'Maggie', 48, 'Dog', 'Chihuahua'),
(85, 'Tiger', 14, 'Dog', 'Rottweiler'),
(86, 'Jay', 39, 'Bird', 'African Grey'),
(87, 'Duke', 17, 'Bird', 'Lovebird'),
(88, 'Casey', 42, 'Dog', 'Jack Russel'),
(89, 'Sam', 43, 'Cat', 'Burmese'),
(90, 'Max', 24, 'Cat', 'Persian'),
(91, 'Lucy', 25, 'Dog', 'Golden Retriever'),
(92, 'Cleo', 47, 'Bird', 'Parakeet'),
(93, 'Jack', 8, 'Bird', 'Toucan'),
(94, 'Sam', 52, 'Dog', 'Pitbull'),
(95, 'Toby', 31, 'Bird', 'Budgie'),
(96, 'Tiger', 33, 'Dog', 'Yorkshire'),
(97, 'Sparky', 40, 'Bird', 'Budgie'),
(98, 'Spike', 42, 'Dog', 'German Shepherd'),
(99, 'Jake', 13, 'Cat', 'Abyssinian'),
(100, 'Spike', 34, 'Dog', 'Boxer'),
(101, 'Fido', 7, 'Cat', 'Burmese'),
(102, 'Murphy', 52, 'Cat', 'Abyssinian'),
(103, 'King', 31, 'Dog', 'Beagle'),
(104, 'Jay', 52, 'Dog', 'Chihuahua'),
(105, 'Max', 21, 'Cat', 'Burmese'),
(106, 'Fido', 39, 'Cat', 'Burmese'),
(107, 'Jake', 7, 'Dog', 'Pitbull'),
(108, 'Shadow', 42, 'Bird', 'African Grey'),
(109, 'Rover', 26, 'Dog', 'Jack Russel');
/*!40000 ALTER TABLE `tblpets` ENABLE KEYS */;

/*!40101 SET SQL_MODE=@OLD_SQL_MODE */;
/*!40014 SET FOREIGN_KEY_CHECKS=@OLD_FOREIGN_KEY_CHECKS */;
/*!40014 SET UNIQUE_CHECKS=@OLD_UNIQUE_CHECKS */;
/*!40101 SET CHARACTER_SET_CLIENT=@OLD_CHARACTER_SET_CLIENT */;
/*!40101 SET CHARACTER_SET_RESULTS=@OLD_CHARACTER_SET_RESULTS */;
/*!40101 SET COLLATION_CONNECTION=@OLD_COLLATION_CONNECTION */;
/*!40101 SET CHARACTER_SET_CLIENT=@OLD_CHARACTER_SET_CLIENT */;

```

File: Pets.mdb**tblOwner**

tblOwners				
OwnerID	LastName	FirstName	Telephone	City
5	Schutte	Monique	074 521 8820	Cape Town
6	Fouche	Emma	082 221 2659	Bloemfontein
7	Schutte	Nicole	072 393 5538	Centurion
8	Ngubane	Tanya	082 658 3903	Polokwane
9	Bezuidenhout	Jana	082 770 2752	Polokwane
10	Davey	Kelly	084 490 3274	Ulundi
11	Theunis	Melissa	082 497 7912	Cape Town
12	Raphael	Courtney	084 816 1243	Polokwane
13	Watts	Michelle	082 617 9619	Richards Bay
14	Adriaans	Simone	084 512 8971	Midrand
15	Ngubane	Amy	084 598 3540	Midrand
16	Way	Nadia	082 377 3329	Pretoria
17	Swanepoel	Nicky	084 584 0536	Johannesburg
18	Swanepoel	Chantel	082 714 1751	Cape Town
19	Grobler	Zinhle	074 299 3948	Kempton Park
20	Du Preez	Caitlin	074 215 7964	Knysna
21	Petzer	Charlene	082 517 9383	Ulundi
22	Duma	Jodi	082 827 3009	Cape Town
23	Brophy	Palesa	074 631 7922	Port Elizabeth
24	Wagener	Lerato	074 897 1259	Ulundi
25	Watts	Jessica	072 691 5542	Centurion
26	Adriaans	Mikaela	083 550 2341	Port Elizabeth
27	West	Christine	083 434 3770	Centurion
28	Williams	Toni	084 689 5760	Durban
29	Bezuidenhout	Daniel	084 456 9284	Midrand
30	Mbeki	Nickens	084 249 0903	Pretoria
31	Theunis	Andrew	074 355 8272	Cape Town
32	Fouche	Kevin	074 491 4832	East London
33	Watts	David	072 391 0409	Durban
34	Kent	Junior	084 391 1222	Cape Town
35	Smit	Patrick	084 644 7473	Port Elizabeth
36	Theunis	Daouda	073 243 1648	Midrand
37	Fouche	Jacques	073 331 0430	Johannesburg
38	Moloto	Jarryd	074 214 5504	Richards Bay
39	Watts	Clinton	072 360 8071	Pietermaritzburg
40	Anfield	Thato	073 788 0908	Centurion
41	West	Peter	072 581 0600	Richards Bay
42	Van Wyk	Jacob	083 473 2686	Cape Town
43	Bezuidenhout	Lebogang	083 874 0060	East London
44	Petzer	Luigi	074 575 5078	Port Elizabeth

tblOwners				
OwnerID	LastName	FirstName	Telephone	City
45	Ngubane	Chris	083 470 8076	Knysna
46	Theunis	Katleho	083 798 4995	Ulundi
47	Culling	Thabo	072 564 6997	Richards Bay
48	Louw	Mervin	082 634 7456	Durban
49	Way	Jonathan	073 382 7666	Ulundi
50	Moloto	Nhlanhla	074 275 6930	Port Elizabeth
51	Hlangani	Gift	073 809 7287	Knysna
52	Way	Johan	083 561 8246	Richards Bay

tblPets

tblPets				
PetID	PetName	OwnerID	PetType	PetBreed
11	Harley	28	Cat	Abyssinian
12	Cleo	11	Bird	African Grey
13	Buster	40	Cat	Persian
14	Donner	11	Bird	Canary
15	Rusty	14	Bird	Cockatoo
16	Maggie	33	Cat	Siamese
17	Snoopy	36	Bird	Parakeet
18	Paul	51	Bird	Toucan
19	Tiger	24	Dog	Doberman
20	Sparky	12	Bird	Budgie
21	King	43	Bird	Cockatoo
22	Casey	34	Bird	Lovebird
23	Rocky	37	Bird	Dove
24	Spot	15	Dog	Boxer
25	Donner	46	Cat	Burmese
26	Shelby	32	Bird	Finch
27	Branford	35	Cat	Burmese
28	King	50	Dog	Beagle
29	Cody	41	Cat	Abyssinian
30	Fluffy	17	Bird	African Grey
31	Toby	46	Bird	Toucan
32	Buster	13	Cat	Burmese
33	Winston	36	Dog	Yorkshire
34	Rex	26	Bird	Cockatoo
35	Branford	30	Dog	Rottweiler
36	Tiger	28	Cat	Burmese
37	Prince	39	Cat	Burmese
38	Molly	40	Dog	Rottweiler
39	Murphy	15	Cat	Burmese
40	Tiger	42	Dog	Rottweiler
41	Charlie	35	Dog	Pitbull

tblPets				
PetID	PetName	OwnerID	PetType	PetBreed
42	Winston	52	Dog	Labrador
43	Sparky	33	Cat	Abyssinian
44	Casey	40	Bird	Canary
45	Cody	23	Cat	Abyssinian
46	Sadie	22	Dog	Chihuahua
47	Branford	33	Dog	Golden Retriever
48	Charlie	46	Bird	Dove
49	Shelby	7	Bird	Lovebird
50	Charlie	21	Cat	Persian
51	Bailey	22	Bird	Parakeet
52	Rover	44	Cat	Burmese
53	Dopey	38	Cat	Burmese
54	Shadow	51	Dog	Doberman
55	Rex	26	Bird	Toucan
56	Sparky	8	Cat	Persian
57	Toby	19	Bird	Dove
58	Charlie	47	Cat	Siamese
59	Jack	44	Cat	Siamese
60	Rex	47	Bird	African Grey
61	Fluffy	16	Dog	Doberman
62	Lucy	31	Dog	Rottweiler
63	Rusty	17	Dog	Doberman
64	Daisy	52	Cat	Siamese
65	Bailey	24	Bird	Lovebird
66	Rocky	25	Bird	Dove
67	Bailey	41	Bird	Canary
68	Tiger	30	Dog	Pitbull
69	Princess	32	Dog	German Shepherd
70	Duke	40	Cat	Persian
71	Princess	15	Dog	Boxer
72	Spike	22	Dog	German Shepherd
73	Harley	37	Bird	Conure
74	Sadie	49	Dog	Yorkshire
75	Paul	6	Bird	Canary
76	Donner	51	Bird	Finch
77	Barney	29	Dog	Jack Russel
78	Cody	17	Bird	Toucan
79	Dopey	39	Dog	Doberman
80	Molly	38	Dog	German Shepherd
81	Spot	38	Cat	Persian
82	Duke	29	Dog	Beagle
83	Rocky	8	Cat	Siamese

tblPets				
PetID	PetName	OwnerID	PetType	PetBreed
84	Maggie	48	Dog	Chihuahua
85	Tiger	14	Dog	Rottweiler
86	Jay	39	Bird	African Grey
87	Duke	17	Bird	Lovebird
88	Casey	42	Dog	Jack Russel
89	Sam	43	Cat	Burmese
90	Max	24	Cat	Persian
91	Lucy	25	Dog	Golden Retriever
92	Cleo	47	Bird	Parakeet
93	Jack	8	Bird	Toucan
94	Sam	52	Dog	Pitbull
95	Toby	31	Bird	Budgie
96	Tiger	33	Dog	Yorkshire
97	Sparky	40	Bird	Budgie
98	Spike	42	Dog	German Shepherd
99	Jake	13	Cat	Abyssinian
100	Spike	34	Dog	Boxer
101	Fido	7	Cat	Burmese
102	Murphy	52	Cat	Abyssinian
103	King	31	Dog	Beagle
104	Jay	52	Dog	Chihuahua
105	Max	21	Cat	Burmese
106	Fido	39	Cat	Burmese
107	Jake	7	Dog	Pitbull
108	Shadow	42	Bird	African Grey
109	Rover	26	Dog	Jack Russel

Student Files: Question 2

File: Delegate.java (Text-based)

```
public class Delegate
{
    protected String firstName;
    protected String lastName;
    protected String company;
    protected String streamCodes;

    public Delegate (String fn, String ln, String co, String sc)
    {
        firstName = fn;
        lastName = ln;
        company = co;
        streamCodes = sc;
    }
}
```

File: Delegate.java (Netbeans-based)

```
package findapet;

public class Delegate {

    protected String firstName;
    protected String lastName;
    protected String company;
    protected String streamCodes;

    public Delegate(String fn, String ln, String co, String sc) {
        firstName = fn;
        lastName = ln;
        company = co;
        streamCodes = sc;
    }
}
```

File: ConventionApp.java (Text-based)

```
import javax.swing.*.*;

public class ConventionApp
{
    public static void main (String[] args)
    {
        char choice;
        do
        {
            choice = getMenuChoice ();

            if (choice == 'A')
            {

            }

            else if (choice == 'B')
            {

            }

        }
    }
}
```

```

        else if (choice == 'C')
        {

        }

        else if (choice == 'D')
        {

        }

    }
    while (choice != 'X');
}

public static char getMenuChoice ()
{
    String menu = "A) List all delegates\nB) List all Gold delegates\nC)
                  Sort delegates\nD) Show the number of
                  delegates attending each seminar\nX) Exit";

    char choice = JOptionPane.showInputDialog (menu).toUpperCase ().charAt
                                (0);

    return choice;
}
}

```

File: ConventionApp.java (Netbeans-based)

```
package findapet;
```

```

public class ConventionApp extends javax.swing.JFrame {

    /** Creates new form ConventionApp */
    public ConventionApp() {
        initComponents();
    }

    /** This method is called from within the constructor to
     * initialize the form.
     * WARNING: Do NOT modify this code. The content of this method is
     * always regenerated by the Form Editor.
     */
    @SuppressWarnings("unchecked")
    // <editor-fold defaultstate="collapsed" desc="Generated Code">
    private void initComponents() {

        jMenuBar1 = new javax.swing.JMenuBar();
        mnuMain = new javax.swing.JMenu();
        mnuDisplayAllDelegates = new javax.swing.JMenuItem();
        mnuDisplayGoldDelegates = new javax.swing.JMenuItem();
        mnuSortDelegates = new javax.swing.JMenuItem();
        mnuPeopleAttendingSeminars = new javax.swing.JMenuItem();

        setDefaultCloseOperation(javax.swing.WindowConstants.EXIT_ON_CLOSE);
        setTitle("IEB IT Prac 2009");

        mnuMain.setText("Menu");

        mnuDisplayAllDelegates.setText("Display all Delegates");
        mnuMain.add(mnuDisplayAllDelegates);
    }
}

```

```

        mnuDisplayGoldDelegates.setText("Display Gold Delegates");
        mnuMain.add(mnuDisplayGoldDelegates);

        mnuSortDelegates.setText("Sort Delegates");
        mnuMain.add(mnuSortDelegates);

        mnuPeopleAttendingSeminars.setText("People Attending Seminars");
        mnuMain.add(mnuPeopleAttendingSeminars);

        jMenuBar1.add(mnuMain);

        setJMenuBar(jMenuBar1);

        javax.swing.GroupLayout layout = new
                                javax.swing.GroupLayout(getContentPane());
        getContentPane().setLayout(layout);
        layout.setHorizontalGroup(

                                layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
                                .addGap(0, 400, Short.MAX_VALUE)
        );
        layout.setVerticalGroup(

                                layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
                                .addGap(0, 279, Short.MAX_VALUE)
        );

        pack();
    } // </editor-fold>

    /**
     * @param args the command line arguments
     */
    public static void main(String args[]) {
        java.awt.EventQueue.invokeLater(new Runnable() {
            public void run() {
                new ConventionApp().setVisible(true);
            }
        });
    }

    // Variables declaration - do not modify
    private javax.swing.JMenuBar jMenuBar1;
    private javax.swing.JMenuItem mnuDisplayAllDelegates;
    private javax.swing.JMenuItem mnuDisplayGoldDelegates;
    private javax.swing.JMenu mnuMain;
    private javax.swing.JMenuItem mnuPeopleAttendingSeminars;
    private javax.swing.JMenuItem mnuSortDelegates;
    // End of variables declaration
}

```

File: uDelegate.pas (Delphi)

```
unit uDelegate;

interface

type Delegate = class(tobject)
    private

        protected
            firstname : string;
            lastname : string;
            company : string;
            streamcodes : string;
        public
            constructor create (fn, ln, c, sc : string);
    end;

implementation

{ Delegate }

constructor Delegate.create(fn, ln, c, sc: string);
begin
    firstname := fn;
    lastname := ln;
    company := c;
    streamcodes := sc;
end;

end.
```

File: ufrmMain.pas (Delphi)

```
unit ufrmMain;

interface

uses
    Windows, Messages, SysUtils, Variants, Classes, Graphics, Controls, Forms,
    Dialogs, Menus;

type
    TForm1 = class(TForm)
        mm1: TMainMenu;
        Options1: TMenuItem;
        Quit1: TMenuItem;
        DisplayAllDelegates1: TMenuItem;
        DisplayGoldDelegates1: TMenuItem;
        SortDelegates1: TMenuItem;
        Displayattendancefigures1: TMenuItem;
        procedure Quit1Click(Sender: TObject);
    private
        { Private declarations }
    public
        { Public declarations }
    end;

var
    Form1: TForm1;

implementation
```

```
{ $R *.dfm }

procedure TForm1.Quit1Click(Sender: TObject);
begin
    Close;
end;

end.
```

File: conference.txt

Linda#Ngubane#Systro#P0
 Patrick#North#Tcsi Technology#OP
 Mervin#Shaw#Gratation#SDOP#912
 Clara#Lanuschagne#Laticrond#PSD0
 Lisa#Mbebe#Systro#OPDS#822
 Alexis#Mondell#Ltria Incorporated#DP#224
 Brandon#Rowe#Systork USA#DP
 Suanne#De Klerk#Systork USA#ODS#408
 Robyn#Miller#VER Design#SP0
 Samira#Govender#Seart Manufacturing#OPSD#805
 Samantha#Moodley#ING Manufacturing#DSP#714
 Kieran#Fearson#Ated Systems#PDS0
 Rizzy#Moodley#Micor#DOSP
 Melissa#Jordan#Seart Manufacturing#DS#402
 Carla#Massey#CES Incorporated#DS0
 Annie#van Zyl#Seart Manufacturing#PSD0#311
 Matthew#Herman#DEM Corporation#OSDP
 Herman#Jacobs#COR Design#PDS#115
 Jessica#le Sueur#Tcsi Technology#DS
 Katleho#Morana#Communductor#DOSP#309
 Kuhle#Mabena#Systro#DS0
 Luarent#Perry#Inted Devices#DOS#624
 Michelle#Millar#Logiesk#DP0
 Vanessa#Wooders#Allects Technology Partners#DOS#605
 David#Milliband#Temicro#DSOP#816
 Cory#Smith#Asce Consulting#DPS0#911
 Junior#Ndleve#Exard Development#SPD
 Jarryd#Wright#Cypresk Design#DS0#519
 Dumisani#Sibongile#Natemicro Design#PSOD
 Carishka#Reddy#Cypresk Design#POSD#217
 Peter#Northern#MAN Systems#SP0#520
 Jonathan#Swift#Hewlech#DS
 Andrew#Keriston#Aterstems PLC#POSD#319
 Moxy#Landman#Micor#POS#718
 Ashton#Freeman#Micor#PS#920
 James#Mason#Adobilogied Development#PSD
 Julian#Cilliers#Seart Manufacturing#PSD0#602
 Trevor#Ferguson#Quant Manufacturing#PSOD
 Mpho#Sithole#CON Europe#D#615
 Simone#Reedling#Conatio Capital#PS0#515
 Toni#Hugo#Acorp PLC#POS
 Nickens#Andrews#Micor#DSOP#906
 Christo#Van Den Berg#Inetwarts Asia#D
 Ntombi#Buthelezi#Systorporpor Investments#PSD
 Jacob#Goldman#Systro#DSOP
 Tamryn#Greevers#IMA Technology Partners#DPS#117
 Jana#Jacobs#COR Design#PS#416
 Garrin#De Nobrega#ING Manufacturing#PDOS#507
 Emma#Malherbe#Imaximagics Devices#PDS0
 Kelly#Mouton#Incommun Systems#SP0
 Johan#Du Toit#Imaximagics Devices#SD0
 Ashlee#Simson#Cypresk Design#PDS#918

Amber#Mandalay#BAS Research#OS#114
Jacques#Cousteau#COR Design#PSD#325
Naudine#Van Wyk#Ated Systems#SPD0#304
Nhlanhla#Duma#Oper#DSOP
Jean#Smith#Seart Manufacturing#POSD
Tammy#Simpson#COR Design#SDOP
Tanya#Meinard#Temicro#SD#102
Chris#Tucker#Systro#ODPS#423
Craig#Steyn#Oper#DP
Remy#Jameson#Valm#DOS#422
Daniel#Nickson#Genced Devices#PDOS
Ruby#O Rayles#Dentury Group#D0#113
Mikaela#Pienaar#Brosys Systems#D
Clinton#Smith#Seart Manufacturing#OPDS#721
Courtney#Walsh#Pachipme Ventures#DP#412
Gift#Mbali#Natemicro Design#DSO
Lebogang#Marogo#Macom Europe#D0#222
Nicole#Malan#INE Group#SOP
Luca#Vettori#Ciscoram#OP
Damien#Levine#Systro#P#417
Aaliyah#Terblanche#Agitec#SODP#908
Johan#Van Der Merwe#Inted#ODPS
Nkosi#Mabuza#Bagilium Europe#DS#806
Thato#Hlangani#Conacts Germany#OSD#120
Wes#Daniels#ING Manufacturing#PSD0
Christine#Retief#ING Manufacturing#PS0#711
Monique#Nel#Ation Capital#POSD
Dinda#Ndebele#Systork USA#DSP#302
Kevin#Davids#INX Japan#DPOS
Melany#Naidoo#Mecuridial Technology#PSD0#321
Megan#Smith#Texxonical#POS
Thabo#Maruma#Imaximagics Devices#SPD#121
Slungile#Khumalo#Natemicro Design#PS0#510
Leyton#Hewit#Alpinted Manufacturing#SD0#409
Luigi#Monticelli#Rate#PDOS#919
Angelique#Roux#Inted Devices#0